

Assessing predictions

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Last time: assessing binary predictions

Thresholding: $\hat{Y}_i = \begin{cases} 1 & \hat{p}_i \geq c \\ 0 & \hat{p}_i < c \end{cases}$

Confusion matrix:

	$Y = 0$	$Y = 1$
$\hat{Y} = 0$	True Negatives (TN)	False Negatives (FN)
$\hat{Y} = 1$	False Positives (FP)	True Positives (TP)

Last time

$$c = 0.33$$

Predicted Status	Actual Status	
	negative	positive
negative	3069	430
positive	954	1267

as $c \uparrow$
sens. $\downarrow$,
Spec. $\uparrow$

$$c = 0.2$$

Predicted Status	Actual Status	
	negative	positive
negative	2327	239
positive	1696	1458

$$c = 0.5$$

Predicted Status	Actual Status	
	negative	positive
negative	3634	765
positive	389	932

Changing thresholds

$c = 0.5$ maximizes accuracy (in population)

$$\text{i.e. if } \hat{p} = p \\ \text{if } \hat{y}_i = \begin{cases} 1 & \text{if } p_i \geq c \\ 0 & \text{if } p_i < c \end{cases}$$

then $c = 0.5$ maximizes $P(\hat{Y} = Y)$
(proof: supplementary notes)

But accuracy is not everything;

- care about other aspects like interpretability
- Accuracy means less when high class imbalance

E.g.

	$Y=0$	$Y=1$
$\hat{Y}=0$	4000	100
$\hat{Y}=1$	0	0

$$\text{sensitivity} = 0$$

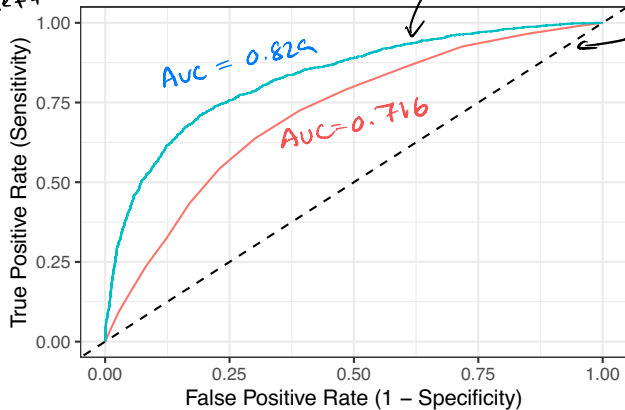
$$\text{Accuracy} = \frac{4000}{4100}$$

Choosing c :

- may be problem dependent
- choose to optimize combo of metrics
 - Sens. + Spec.
 - $\sqrt{\text{sens} \times \text{spec.}}$ or $\sqrt{\text{sens} \times \text{PAV}}$
 - $F_1 = \frac{2}{\text{sens}^{-1} + \text{spec}^{-1}}$

ROC Curves

went close to top left



each point corresponds to a different threshold

random classifier
assign $\hat{y} = 1$
w/ prob p
 $\hat{y} = 0$ w/ prob $1-p$

model

— Age only
— Age, WBC, PLT

$\Rightarrow \text{Sens} = p$

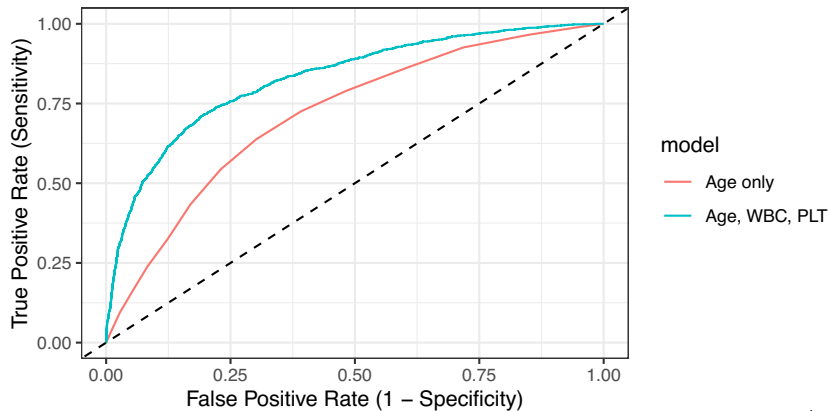
$\text{Spec} = 1-p$

$\Rightarrow 1 - \text{Spec} = p$

(plot over all values of p)

Plot tradeoff between sensitivity & specificity for all thresholds
ideally: could get threshold where $\text{sens.} = 1$ & $\text{spec} = 1$
($1 - \text{spec} = 0$)
Summarizing: area under curve (AUC) $\in [0.5, 1]$
(closer to 1 are better)

ROC Curves



AUC interpretation: probability that a randomly selected observation from class $Y=1$ is ranked higher (i.e. larger $\hat{p}$) than a randomly selected observation from class $Y=0$

Another example: loan defaults

Data on 948 real estate loans from CA, which were guaranteed by the US Small Business Administration (SBA). Variables include:

- ▶ **Default**: whether the business defaulted on the loan (1 = yes, 0 = no)
- ▶ **UrbanRural**: 1 if business is in urban area, 2 if business is in rural area, 0 if unknown
- ▶ **NewExist**: 1 if business already existed, 2 if business is new, 0 if unknown
- ▶ **Term**: Length of the loan term (months)
- ▶ **NoEmp**: Number of employees of the business before receiving the loan
- ▶ **GrAppv**: Gross amount of loan approved by the bank, in dollars
- ▶ **SBA_Appv**: Amount of the loan guaranteed by the SBA, in dollars
- ▶ **DisbursementGross**: The amount of money disbursed (loaned), in dollars

Another example: loan default

Fit a model on the 948 loans:

	Estimate	Std. Error	z value	$\Pr(> z)$
(Intercept)	6.225	1.055	5.899	0.000
$\log(\text{GrAppv})$	-0.154	0.101	-1.521	0.128
$\log(\text{NoEmp} + 1)$	0.059	0.126	0.466	0.641
Term	-0.074	0.005	-15.210	0.000

Default? (predicted)	Default? (actual)	
	no	yes
no	438	82
yes	92	336

Predicting on new data?

Predictions on the original 948 loans used to fit the model:

Default? (predicted)	Default? (actual)	
	no	yes
no	438	82
yes	92	336

$$\text{deviance} = -2 \sum_{i=1}^n \log f(y_i | \hat{p}_i)$$

$$\frac{836.06}{948} = 0.882$$

Acc. = 0.816, Sens. = 0.804, Spec. = 0.826, Deviance = 836.06

Predictions on a **new** set of 407 loans (not used to fit model):

Default? (predicted)	Default? (actual)	
	no	yes
no	165	53
yes	44	145

$$\frac{405.78}{407} = 0.997$$

Acc. = 0.762, Sens. = 0.732, Spec. = 0.790, Deviance = 405.78

Predicting on new data

In general: A model predicts **better** on the data used to fit the model than it does on **new** data.

Why?

Because the fitted model is
optimized for the original data used
(minimizing SSE, maximizing likelihood, etc.)

Training data: original data used to fit model

Test data: new (unseen) data, not used to
fit model

Class activity

Work on the class activity (course website)